

THOMSEN RIVER, NT, Canada

WMO: 719160

Lat: 73.2311N

Lon: 119.5406W

Elev: 42

StdP: 100.83

Time Zone: -7.00 (NAM)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-44.7	-43.1	-49.6	0.0	-44.7	-47.8	0.0	-43.1	15.8	-24.9	14.0	-24.7	1.0	150	1.116

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	18.9	13.1	16.4	11.9	13.9	10.5	13.7	18.2	12.2	15.8	10.6	13.6	5.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	8.3	15.3	9.9	7.6	13.7	8.5	6.9	12.2	38.5	18.2	34.6	16.0	30.9	13.7	19.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	12.2	10.8	DB	-46.3	21.2	2.3	3.1	-47.9	23.5	-49.2	25.3	-50.5	27.0	-52.2	29.3	(4)
(5)			WB	-46.3	15.1	2.2	2.0	-47.9	16.5	-49.2	17.6	-50.5	18.7	-52.1	20.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-14.0	-31.7	-32.0	-29.4	-20.6	-7.9	4.1	8.6	5.4	-1.7	-12.6	-22.5	-28.8		
	DBStd	15.84	6.43	6.84	7.00	6.19	5.49	4.30	4.16	4.01	4.37	5.95	6.41	6.95		
	HDD10.0	8809	1292	1175	1222	919	555	184	79	154	350	702	974	1203		
	HDD18.3	11799	1550	1408	1481	1169	813	427	302	402	600	960	1224	1461		
	CDD10.0	53	0	0	0	0	0	7	36	11	0	0	0	0		
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	4.7	4.3	4.0	4.2	4.5	4.8	5.3	5.5	4.9	5.2	4.7	4.3	4.2		
(15) Precipitation	PrecAvg	151	6	6	7	8	10	13	27	28	19	14	9	6		
	PrecMax	206	15	22	15	21	25	30	47	55	35	37	22	26		
	PrecMin	89	1	1	1	1	2	2	9	6	5	2	1	1		
	PrecStd	34	3	4	3	5	6	7	11	12	8	8	5	5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-12.7	-11.2	-8.8	-0.8	4.5	18.6	22.6	21.1	10.0	0.7	-3.6	-9.7		
		MCWB	-12.8	-11.5	-9.1	-1.4	2.6	12.8	14.4	15.3	7.7	0.7	-3.8	-10.0		
	2%	DB	-16.7	-15.3	-12.9	-6.1	2.9	15.2	19.9	17.2	7.5	-1.0	-7.7	-13.1		
		MCWB	-17.0	-15.7	-13.3	-6.7	1.6	11.2	13.4	13.3	6.5	-1.2	-8.0	-13.4		
	5%	DB	-19.6	-18.3	-16.3	-9.0	1.2	12.7	17.9	13.7	5.8	-2.7	-10.9	-16.0		
		MCWB	-19.8	-18.5	-16.6	-9.5	0.4	9.4	12.3	11.1	5.2	-3.0	-11.2	-16.3		
	10%	DB	-21.9	-21.0	-19.2	-11.2	-0.2	10.4	15.7	11.2	4.1	-4.3	-13.2	-18.5		
		MCWB	-22.1	-21.2	-19.5	-11.7	-1.0	7.9	11.3	9.2	3.7	-4.6	-13.5	-18.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-12.9	-11.6	-9.0	-1.5	3.1	13.5	15.1	15.8	8.4	0.6	-3.8	-9.9		
		MCDB	-12.7	-11.3	-8.8	-0.8	4.3	17.7	20.8	20.2	9.2	0.6	-3.6	-9.8		
	2%	WB	-17.0	-15.6	-13.3	-6.6	1.8	11.6	13.9	13.6	6.6	-1.4	-8.0	-13.4		
		MCDB	-16.8	-15.3	-12.9	-6.1	2.7	14.7	19.2	17.0	7.4	-1.3	-7.7	-13.2		
	5%	WB	-19.8	-18.5	-16.6	-9.4	0.4	9.7	12.7	11.3	5.2	-3.1	-11.2	-16.3		
		MCDB	-19.6	-18.2	-16.3	-9.0	1.2	12.4	17.2	13.3	5.7	-2.9	-10.9	-16.0		
	10%	WB	-22.2	-21.2	-19.5	-11.7	-1.0	8.0	11.5	9.4	3.7	-4.6	-13.5	-18.7		
		MCDB	-21.9	-21.0	-19.2	-11.2	-0.3	10.3	15.2	11.0	4.0	-4.3	-13.2	-18.5		
(35) Mean Daily Temperature Range	MDBR		7.4	7.1	7.9	9.5	7.2	5.7	6.9	6.1	4.4	6.4	7.4	7.1		
	5% DB	MCDBR	10.5	9.5	8.4	9.3	7.2	8.8	9.4	9.8	5.8	5.8	9.0	9.4		
		MCWBR	10.4	9.4	8.2	8.8	5.9	5.3	4.5	5.9	4.7	5.9	9.0	9.3		
	5% WB	MCDBR	10.5	9.3	8.5	9.1	6.8	8.3	9.3	9.2	5.4	5.9	9.0	9.3		
		MCWBR	10.4	9.2	8.3	8.7	5.6	5.4	5.1	5.7	4.5	5.9	9.0	9.3		
(40) Clear-Sky Solar Irradiance	taub		N/A	0.177	0.244	0.316	0.331	0.334	0.342	0.331	0.280	0.173	N/A	N/A		
	taud		N/A	1.901	2.152	2.017	2.093	2.374	2.420	2.433	2.397	1.928	N/A	N/A		
	Ebn at Noon		0	389	714	765	817	841	817	773	688	367	0	0		
	Edh at Noon		0	26	63	111	120	95	88	75	53	23	0	0		
(44) All-Sky Solar Radiation	RadAvg		0.00	0.17	1.46	4.14	6.05	5.87	4.96	2.95	1.16	0.31	0.00	0.00		
	RadStd		0.00	0.02	0.08	0.19	0.44	0.55	0.46	0.35	0.12	0.05	0.00	0.00		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page